

Contents

1 · The mandate	2
2 · The reversal group	4
3 · The criterion and the normal form	5
4 · The laundering theorem beyond the ring	6
5 · The genericity and the stratification	7
6 · The binary resolution floor	8
7 · The shadow theorem	9
8 · The exact free-cell machinery	10
9 · The free scan: the escape room is empty	11
10 · The line atom: the cell's $D(2)$ solved	13
11 · The certificate ledger	14
12 · The verdict and the open ledger	15

1 · The mandate

The previous ledger closed with two frontiers. Volume X had generalized the rank-defect theorem and discovered, in its battery's exact zeros, the reflection-laundrying theorem: if a state-space symmetry conjugates a chain's dynamics to its own time reversal, every symmetry-invariant sensor launders the arrow exactly, and on the N-ring every reflection is such a symmetry — the ring's arrow is its chirality. What remained open was the theorem's life beyond the ring: which irreducible chains admit a reversal symmetry at all, what the symmetry structure is, and which sensors preserve the arrow off the circulant class. The second frontier was Volume IX's honest interval for the (1,1) cell — the sandwich distance $D(2)$ bounded between the EYM floor 1 and the measured 1.2771427 — together with Open 7.13, the multiletter Hankel equality off the level-constant class, now framed by the abelian-shadow theorem: the rank defect is the abelian shadow of the arrow, and the abelianized rung is the shadow of the free problem.

This volume discharges both. The first frontier becomes the reversal group: a two-coset extension of the automorphism group inside the symmetric group, with a computable criterion (the anti-centralizer form), a normal form that manufactures non-circulant witnesses at will, an exact dimension table that makes the whole reflection mechanism measure-zero, and a stratification of every laundrying route — with Volume X's doubly-stochastic witness adjudicated: its laundrying is not reflection at all, and the sharp converse is false. Along the way the battery discovered a theorem cluster nobody had ordered: the binary resolution floor, in which the 2-block and 3-block laws of every stationary binary process are symmetric by identity, and the first possible binary arrow is the 4-block law. The second frontier becomes the shadow sandwich: an exact free-cell machinery with closed-form Grams, a free scan that shows the non-commutative 2-state class does not escape the abelian shadow, and then the unexpected closing move — the cell's optimum runs to the boundary of the approximant variety, the line-atom family, where the whole problem collapses to a two-parameter cubic stationarity system whose solution is the value of $D(2)$ to forty digits.

The ordered link	The deliverable	Status	Verification
(1) The reflection-laundering theorem beyond the ring — which irreducible chains admit a reversal symmetry; the arrow-preserving sensor lattice off the circulant class	The reversal group $\hat{G} = \text{Aut} \sim R$ (index-2 normal coset structure; asymmetric chains have exactly one reversal symmetry, an involution); the anti-centralizer criterion $R(P) = \text{centralizer}(S)$ intersect anti-centralizer(K); the twist normal form $F_{ij} = F_{\{\rho(j)\rho(i)\}}$ with manufactured non-circulant witnesses; the laundering theorem verified beyond the ring; the exact dimension table (the strata are proper linear varieties, measure zero); the laundering stratification (reflection / strong lumping / residual) with the DS witness adjudicated; the binary resolution floor theorem package	PROVED (the group laws, the criterion, the normal form, the laundering, the genericity, the binary identities) and WITNESSED (the sharp converse FALSE: the strong-lumping route)	reversal_group.py: the ring's $\hat{G} = D_N$ for $N = 3..8$; the criterion 120/120; pi-preservation $5.6e-17$; the twist error 0.0 on four witnesses; the laundering zeros exact at $n = 2..6$; the dimension table by exact integer row reduction; 0/3000 random self-converse; the binary identities to $1.1e-16$
(2) Vol IX's still-open cell $D(2)$ / Open 7.13 — now framed by the abelian-shadow theorem	The shadow theorem: the dilation identity ($\sigma(\text{free}) = \sigma(\text{shadow})$ exactly) and the lift ($D_{\text{free}} \leq D_{\text{abelian}}$): the sandwich $1 \leq D_{\text{free}}(2) \leq D_{\text{abelian}}(2)$; the exact free-cell machinery (the 6x6 closed-form eigenproblem with the Lyapunov identities); the free scan over the 2-state WFAs — the escape room is empty, so the off-class equality fails on the witness at the measured level; and the cell's $D(2)$ SOLVED: the optimum is the line-atom boundary family, reduced to a 2-parameter cubic stationarity system solved to 40 digits	SOLVED: $D(2) = 1.2771421129\dots$ (the line-atom stationary value; six digits stable across three independent optimizers, then 50 digits by the algebraic solve); the free-side verdict MEASURED; the minimal polynomial left open with the artifacts saved	free_cell.py: the machinery validated three ways (zero exact, abelian referee $1.2e-9$, dense truncation $1.7e-7$); the diagonal subscan reproduces the Vol IX optimum (the lift verified); line_atom.py: the closed 6x6 validated exact, the gamma_1-parity block decoupling, the cubic stationarity solved to 90 digits

Table 1 — The order, itemized and answered: the two frontiers named at the close of the previous ledger (Task 15's remaining links), the theorem package or construction that discharges each, its status, and its verification in this session's batteries

The standing honesty discipline applies throughout. Every number below is either computed exactly in this session's batteries (reversal_group.py, free_cell.py, line_atom.py, minpoly_cell2.py, each with a results JSON) or labelled with its precision. Three mid-session defects were caught by the anchor discipline and are recorded because they shaped the claims: a word-encoding bug (the block word lists stored one-letter tuples, so the word 'ab' was read as a single unknown letter) that made the free machinery disagree with a dense truncation referee until fixed; a witness-rounding bug that briefly destroyed the exactness of the laundering zeros; and an elimination-chain error in the minimal-polynomial computation (eliminating c before y) that produced a bivariate impostor. Each was exposed by an independent check, and the batteries were re-run clean before any claim was read off.

2 · The reversal group

Fix an irreducible stochastic matrix P with stationary distribution π , its time reversal $\hat{P} = \pi^{-1} P^T \pi$, the automorphism group $\text{Aut}(P)$ of permutation conjugations, and the reversal symmetries $R(P) = \{\rho : \rho P \rho^{-1} = \hat{P}\}$. The first theorem is that these objects assemble into a group: π -preservation comes free (every reversal symmetry preserves the stationary measure, verified to $5.6e-17$ across the battery), the products close (R times R lands in Aut , Aut times R stays in R), and when R is nonempty the automorphism group is an index-2 normal subgroup of the reversal group $\hat{G} = \text{Aut} + R$, so that R is a single coset — $\rho_0 \text{Aut} = \text{Aut} \rho_0$ — and the square of every reversal symmetry is an automorphism. The corollary is the structure of chirality: if the chain is asymmetric (Aut trivial) and a reversal symmetry exists at all, it is unique and it is an involution. The handedness of an asymmetric driven chain is a single coin, not a family.

The driven ring is the canonical example and the theorem's connection to Volume X. For the (a,b) -ring with a different from b , the automorphisms are exactly the rotations Z_N (the reflections conjugate the circulant to its transpose, which is the (b,a) ring — not itself unless $a = b$), and the reversal symmetries are exactly the N reflections: the reversal group is the dihedral group, and Volume X's slogan — the ring's arrow is its chirality — is now the coset statement $R = D_N \setminus Z_N$. At equilibrium the ring is symmetric, Aut swallows the reflections, and $R = \text{Aut}$: the equivalence 'reversible if and only if $R = \text{Aut}$ ' is the group-theoretic restatement of detailed balance. The battery verifies the classification for $N = 3$ through 8 exactly, and the normal-form witnesses of the next section realize the asymmetric corner: four non-circulant driven chains, each with Aut trivial and R a single involution.

Object	$\text{Aut}(P)$	$R(P)$	$\hat{G}$	The fact
The driven (a,b) -ring, $N = 3..8$	Z_N (the rotations)	the N reflections	the dihedral D_N	Vol X's 'the arrow is the chirality' is the coset statement $R = D_N \setminus Z_N$; at equilibrium ($a = b$) P is symmetric and $R = \text{Aut}$ (reversible iff $R = \text{Aut}$)
The normal-form witnesses ($N = 4, 4, 5, 6$)	trivial (asymmetric)	exactly one involution ρ	Z_2	the asymmetric chain has at most one reversal symmetry and it is an involution ($\rho^2 = \text{id}$); the chirality is a single coin flip, verified on all four witnesses
A generic chain (any $N \geq 3$)	generically trivial	empty	trivial	the twist loci are proper linear strata (exact dimension table); the set of chains admitting ANY reversal symmetry is measure zero — 0 hits in 3000 random 4-state chains
$N = 2$ chains	S_2	$= \text{Aut}$	S_2	every 2-state chain is reversible: the reversal group is degenerate; the arrow needs $N \geq 3$ states

Table 2 — The reversal group: the structure facts and the canonical examples. $\text{Aut}(P)$ is the permutation automorphism group; $R(P)$ the reversal symmetries; $\hat{G} = \text{Aut} + R$ the reversal group

The group law is worth one more remark because it explains why the reflection mechanism feels so structured on the ring and so fragile off it. An index-2 normal subgroup is normal by counting alone: there is exactly one nontrivial coset, and every element outside it conjugates the subgroup to itself. The reversal symmetries therefore form a torsor for Aut — a homogeneous space of handedness flips — and the arrow, which is nonzero exactly when R is disjoint from Aut , is measured by the pair $(\hat{G}, \text{Aut})$ up to that torsor. On the ring the torsor is large and abelian (the rotations), which is why so many sensors launder there; on an asymmetric chain the torsor is a single point, which is why only the exactly ρ -invariant sensors launder. The classification of which chains admit reversal symmetries is thus a classification of subgroups of S_N containing an index-2 copy of their own conjugation structure — a finite, computable problem, and the next two sections solve its computable and geometric halves.

3 · The criterion and the normal form

The computable half is the anti-centralizer criterion. Conjugate the dynamics by the square root of the stationary measure, $A = \Pi^{(1/2)} P \Pi^{(-1/2)}$, and split it into its symmetric and skew parts $S = (A + A^T)/2$ and $K = (A - A^T)/2$. A permutation ρ is a reversal symmetry if and only if it commutes with the symmetric part and anti-commutes with the skew part: $R(P) = \text{centralizer}(S) \cap \text{anti-centralizer}(K)$, and $\text{Aut}(P) = \text{centralizer}(S) \cap \text{centralizer}(K)$. The reading is physical: the reversible half of the dynamics must be invariant under the symmetry, and the irreversible half — the skew flux, the arrow's carrier — must be odd under it. A reversal symmetry is exactly a symmetry under which the arrow is odd; the battery verified the criterion against exhaustive search on 120 random chains with perfect agreement, for both R and Aut . The criterion makes every question of this section decidable by finite permutation search: given P , the whole reversal group is a closed-form computation.

The geometric half is the twist normal form. Work in flux coordinates $F = \Pi P$, so that the chain is the matrix of stationary flows with row and column sums equal to π . Detailed balance is the symmetry $F_{ij} = F_{ji}$; the self-converse condition with symmetry ρ is its twisted form $F_{ij} = F_{\{\rho(j) \rho(i)\}}$ — the flux from i to j equals the flux from $\rho(j)$ to $\rho(i)$. The normal form follows: take any π -flow G (rows and columns summing to π , with ρ preserving π) and ρ -symmetrize, $F = (G + \rho G)/2$ where ρG is the twisted transpose: the result is a self-converse π -flow whose chain is stochastic, irreducible, generically driven, and — crucially — not circulant. The battery manufactures four such witnesses ($N = 4, 4, 5$ and 6), each doubly stochastic, driven (D between 0.115 and 0.140), with the twist identity verified to 0.0 and the reversal group exactly the predicted single involution. The self-converse world is strictly larger than the circulant world, and its members are as easy to construct as a Sinkhorn projection: the reflection-laundering theorem finally has a home beyond the ring.

The two halves combine into the classification answer. Which irreducible chains admit a reversal symmetry? Exactly those whose flux satisfies one of the finitely many twist systems $F_{ij} = F_{\{\rho(j) \rho(i)\}}$; the anti-centralizer criterion decides membership in closed form; and the normal form generates

every member from an arbitrary flow. The honest scope note is that the flux twist is LINEAR: the self-converse chains form linear strata of the chain space, not a generic class — the subject of the genericity theorem two sections ahead, where the exact dimension table shows just how thin the strata are. But thin is not empty: within each stratum the arrow is real (the witnesses are driven), the laundering is exact, and the chirality is a single involution — the structure the ring taught us, transplanted into chains that no Fourier mode ever visits.

4 · The laundering theorem beyond the ring

The laundering theorem itself needed no repair, only a world to live in. Statement: if ρ is a reversal symmetry of P and the sensor s satisfies $s(\rho i) = s(i)$, then the observed process is reversible — every block reversal KL is exactly zero. The proof is a pushforward: the chain conjugated by ρ is a stationary Markov chain with transition $\rho P \rho^{-1} = \hat{P}$, so the ρ -image of the forward block law is the reversed block law; a ρ -blind sensor cannot distinguish the image from the original, so the observed block law equals its own reversal. On the ring this was a fact about reflections; on the normal-form witnesses it is a fact about chains with no spectral symmetry whatsoever — the battery verifies it on all four witnesses, at block lengths two through six, with the invariant sensors at machine zero (the largest deviation across all witnesses and lengths is $2.8e-17$).

The arrow-preserving sensor lattice off the circulant class is then a measurement, and Figure 1(a) shows it on the $N = 4$ witness with $\rho = (01)(23)$: fifteen sensors, two of them ρ -invariant (the partitions that merge the orbit pairs), the rest breaking ρ . The invariant two launder exactly. Of the thirteen breaking sensors, nine keep a strictly positive arrow — the observed block KL grows with the block length — and four launder despite breaking ρ : they have wandered into the other strata of the laundering stratification, the strong-lumping conditions or the residual accidents, which the next section maps. The $N = 5$ witness tells the same story at scale (five invariant sensors of fifty-two, forty-three of the breaking sensors with positive arrow). The lattice is the honest answer to the ledger's question: off the circulant class, the arrow survives exactly the sensors that break every reversal symmetry AND miss the other laundering strata — the arrow-preserving sensors are the complement of a union of strata, and on the asymmetric witnesses that complement is large and easy to stay inside.

One witness deserves its own paragraph because it closes a loop with Volume X's parallel chat. The iid-pair process — the binary process built from a four-state chain with one-way edges, whose pairs $(0,1)$ and $(1,0)$ are drawn independently at random phases — launders an infinite chain-level arrow (the one-way edges make the reversal divergence diverge) into a record with zero block KL at every length. The battery's first reading called this a residual accident, a symmetry of the record with no chain-level counterpart; the group theory corrected the reading: the chain's reversal group is nonempty (two symmetries), and the sensor IS invariant under $\rho = (AD)(BC)$ — the phase symmetry that looked like a property of the record is exactly the chain-level reversal symmetry. The laundering is the reflection route

after all, and what made it look accidental is that the symmetry mixes the pair content with the phase. The correction is the theorem earning its keep: the group classifies the laundering, and the classification caught the battery's own first impression.

5 · The genericity and the stratification

How large is the self-converse world? The exact dimension table answers: for each permutation ρ , the twist locus L_ρ — the chains whose flux satisfies the ρ -twist — is cut out by linear equations, and the battery computes the dimensions by exact integer row reduction in flux coordinates. On the 6-dimensional space of 3-state chains: the reversible locus ($\rho = \text{identity}$) has dimension 5 (codimension 1), each transposition locus dimension 4 (codimension 2), the 3-cycle loci dimension 2. On the 12-dimensional 4-state chains: the reversible locus codimension 3, the involution loci codimension 4 to 5, the order-3 and order-4 loci thinner still. Every nontrivial locus is proper: the union over the finitely many ρ is a measure-zero subset of the chain space, and the sampling confirms it — zero self-converse chains in three thousand random 4-state draws. The reflection mechanism is the abelian world's mechanism: it lives on the circulant slice and its nearest relatives, and a generic chain's arrow cannot be laundered by reflection because there is nothing to reflect through.

The stratum	The mechanism	The geometry	The witnesses
Reflection	the sensor is ρ -invariant for a reversal symmetry ρ in $R(P)$: the reversed chain is the ρ -conjugate and the sensor is ρ -blind	the twist loci L_ρ : linear, codim 3 ($N = 3$, transpositions) up to codim 8-9 ($N = 4$, involutions); measure zero	all four normal-form witnesses (the laundering exact at every block length); the iid-pair process (the phase symmetry IS the chain-level reversal symmetry $\rho = (AD)(BC)$)
Strong lumping	the Kemeny-Snell conditions: states in a block have identical block profiles, so the observed process degenerates to a lower-dimensional Markov chain, reversible when the lump is 2-state	linear, codim = merged-profile count; the DOMINANT laundering stratum (codim 1 against codim 3 on 3-state chains)	Vol X's doubly-stochastic hidden-arrow witness: $R = \text{emptyset}$ (verified by exhaustive search over S_3), yet the sensor $\{0\} \{1,2\}$ launders completely — the sharp converse is FALSE; the constructed 4-state strong-lumping witness ($D = 0.334$, laundered)
Residual	a symmetry of the record with no chain-level counterpart: the observed process's own degeneracy	algebraic, measure zero (predicted)	no witness isolated this session (the breaking-but-laundered sensors found on the witness lattices are the candidates; the typing is left open)

Table 3 — The laundering stratification: the routes by which a sensor destroys the chain's arrow, their geometric nature, and the session's witnesses. The ambient is the chain space of dimension $N(N-1)$

The stratification sharpens the sharp converse. If a sensor launders the arrow, must the chain carry a reversal symmetry that the sensor respects? No: Volume X's doubly-stochastic hidden-arrow witness is adjudicated by exhaustive search — its reversal group is empty (all six permutations tested) — yet its sensor $\{0\}|\{1,2\}$ launders completely, at every block length, to machine zero. The route is strong lumping:

states one and two have identical block profiles ($P_{10} = P_{20} = 1/5$, exact in fractions), the Kemeny-Snell conditions hold, and the observed process IS the two-state chain $[[.6,.4],[.2,.8]]$, which is reversible because every two-state chain is. The laundering stratification has three tiers: the reflection strata (linear, thin, the symmetric mechanism), the strong-lumping strata (linear, codimension equal to the merged-profile count — the DOMINANT mechanism, an order of magnitude thicker than reflection on three-state chains), and the residual strata (algebraic, measure zero, no witness isolated this session — the breaking-but-laundered sensors of the witness lattices are the candidates, left open). The one-way valve of Volume X is untouched: no stratum creates an arrow, every stratum only destroys it, and the destruction routes are now all named.

6 · The binary resolution floor

The battery's most unlooked-for discovery began as a debugging surprise: the two-block reversal KL of a binary observation refused to be positive no matter how driven the chain. The reason is an identity: for every stationary binary process, $q(01) = q(10)$ exactly, because the pathwise difference of up- and down-transition counts is bounded by one while stationarity equalizes the rates — the telescoping identity, verified on four hundred random hidden Markov binaries to $1.1e-16$. A pairwise arrow needs at least three symbols. And the surprise deepens at the next block: the 3-block law of every stationary binary process is also symmetric, by the run-counting identity — the reversal of a 3-block pattern is the same run-configuration (011 and 110 both count 1-runs of length at least two, 001 and 100 both count 0-runs of length at least two, the rest are palindromes), so the run statistics that determine the block law determine it reversally too. The binary record cannot carry an arrow below the 4-block: the arrow's resolution has a floor built into the alphabet, not into the dynamics.

What does the 4-block see? The adjacent run-pair order statistics — the rate of a 1-run of length a followed by a 0-run of length b , against the reverse. The binary process is reversible if and only if its run-length sequence is order-reversible. From there the singleton-block theorem closes the three-state world: if either block of the lump is a single state, the binary observation of ANY chain is reversible, because a singleton routes memorylessly — its exit distribution cannot carry the previous run — so the other block's entry sequence is iid and the observed process is an alternating renewal with iid runs. Every binary function of a 3-state chain is reversible (the forced 1+2 split), verified at every block length to $3.0e-17$; the same holds for every $1+(N-1)$ split of every N , verified to $3.3e-17$. The first binary arrow needs both blocks of size at least two: hidden rank four with a 2+2 split, where the battery measures 99.8 percent of random instances carrying a positive 4-block arrow (maximum KL 0.0601), growing with the block length — Figure 1(b) shows the curve rising from exactly zero at $n = 2$ and $n = 3$.

The floor is counted twice, and the two counts agree with Volume X's minimality. The pairwise arrow needs three symbols AND at least three states: the driven 3-ring's identity sensor is the minimal pairwise arrow carrier, and all three of its binary sensors are reflection-orbit sensors that launder — the ring's

arrow survives only at full resolution. The block-pair arrow needs a 2+2 binary lump: hidden rank four, both blocks non-singleton. Between them the floor explains, structurally rather than by reflection alone, why the 3-ring's non-identity sensors all launder: a three-state chain has no binary observation capable of an arrow at any block length, so the laundering of its binary sensors is forced before the group theory even enters. The minimal chiral record of Volume X is minimal in a stronger sense than stated there: it is the resolution floor of the arrow itself.

7 · The shadow theorem

Turn to the cell. The free monoid on two letters; the free cell symbol $h(w) = 1[\text{Parikh}(w) = (1,1)]$ — the indicator of the words ab and ba ; its free Hankel $H(u,v) = h(uv)$ on $\mathbb{I}^2$ of the word space, rank four; and its abelianized shadow, the weighted catalecticant of the same indicator on $\mathbb{I}^2$ of the two-dimensional integer lattice — Volume VIII's ab/ba cell, Volume IX's interval. The first theorem of the second front is the dilation identity: the free symbol IS the abelian symbol composed with the Parikh map, and the block-constant subspace of the word space (spanned by the normalized Parikh-block indicators V) carries the entire free Hankel: $H = V \text{ Cat } V\text{-star}$ with a zero block. Singular values are preserved exactly, so the free cell and its shadow share the spectrum $(\sqrt{2}, \sqrt{2}, 1, 1)$, and the EYM floor at $M = 2$ is $\sigma_3 = 1$ on both sides. The battery verifies the identity by dense computation over the five-word support — the singular values match to $1e-12$ — and the check is the hinge of everything that follows: the free problem and the abelian problem measure the same operator until the approximants differ.

The approximants differ exactly at the commutative boundary, and the second theorem — the lift — measures which side has the advantage. Every abelian rank- M approximant lifts to a free rank- M approximant: a Prony atom $\psi(\gamma) = p \lambda^\gamma$ lifts to the multiplicative symbol $g(w) = p \lambda^{m(w)}$, whose free Hankel is $p \times x\text{-star}$ with $x(w) = \lambda^{m(w)}$ — rank one; the two-atom family is the diagonal two-state weighted automata; the affine family is the shear automata. Therefore the free distance is bounded above by the abelianized distance, and both are bounded below by the common floor: $\sigma_3 = 1 \leq D_{\text{free}}(2) \leq D_{\text{abelian}}(2)$ in $[1, 1.2771427]$. This is the abelian-shadow theorem's sandwich form, and it is Open 7.13's frame: the shadow bounds the free problem from above, the EYM floor from below, and the free question is whether the non-commutative freedom of the two-state class — the letters' matrices that do not commute — buys any of the gap between 1 and 1.2771427. The next three sections build the machinery to answer, answer, and then close the problem in a direction nobody had drawn.

A note on why the free side could plausibly win. The unconstrained rank-2 approximation of the cell's Hankel — dropping the Hankel structure entirely — attains the floor $\sigma_3 = 1$ by the Eckart-Young argument. The price of structure is the price of shift-consistency: the approximant's row functions must satisfy $X(ua) = X(u) A_a$, a two-dimensional linear-recurrence constraint that the Eckart-Young truncation has no reason to respect. In the one-letter case the classical AAK theory shows the price is

zero — the shift-consistent truncation exists and attains the floor; Volume VII's level-constant transport proved the same for the commutative multiletter class. The cell is the boundary object: commutative as a target, off the level-constant class as an approximation problem, and the question is what the two-state non-commutative shift-consistency costs against it.

8 · The exact free-cell machinery

The machinery that answers is a six-by-six eigenproblem with closed-form entries. Take any two-state weighted automaton $g(w) = B A_w C$, with A_a and A_b two-by-two and B, C the boundary vectors; the error operator $M = H_{\text{cell}} - H_g$ has its column space inside the span of the four Parikh-block indicators and the two reachable functions $f_i(u) = (B A_u)_i$, so its squared norm is the largest eigenvalue of the product $C G$ of two six-by-six matrices: G the Gram of that basis, C the coefficient Gram summed over all words. Every entry is closed form: the block sums are finite ($C, A_a C, A_b C$ and the symmetric combination on the four blocks), and the reachable and co-reachable Grams are discrete Lyapunov identities — the matrix inverses $(I - A_a \text{ kron } A_a - A_b \text{ kron } A_b)^{-1}$ — which are the free analogues of Volume IX's multinomial generating function $1/(1 - \cdot)$. No truncation, no tail bounds: every norm the battery reports is the true infinite-operator norm, computed in microseconds.

The validation chain is threefold, and it earned its keep by catching a real bug. First the zero approximant: the machinery reproduces $\|H_{\text{cell}}\| = \sqrt{2}$ exactly. Second the abelian referee: on the diagonal automata the machinery must agree with a completely independent algorithm — a dense box truncation of the catalecticant over the two-dimensional lattice — and it does, to $1.2e-9$ across the test atoms. Third the dense word-space truncation: for random stable automata the exact norms match the top singular value of the dense error matrix over all words up to length nine, to the tail scale (the worst relative gap $1.7e-7$). The bug the chain caught: the block word lists were stored as one-letter tuples, so the word 'ab' was read as a single unknown letter and routed through the wrong matrix product — the machinery disagreed with the referee by twenty percent until the encoding was fixed, and the first version of the free scan had silently converged to a wrong value. The anchor discipline held: the claim was never read off the broken version.

One more structural fact falls out of the machinery for free, and it becomes the closing move of the volume. The reachable functions and the block indicators interact through the gamma-one parity: the whole six-by-six problem decomposes, exactly, into the two parity classes — the blocks with even first coordinate together with the even line function, and the odd blocks with the odd line function. The battery does not use this until the line-atom section, but it is already visible in the matrices: the cell's kernel pairs blocks whose Parikh vectors sum to $(1,1)$, which is an odd condition, and the parity is the same one that defined the binary resolution floor's run structure. The two fronts of this volume share a parity — the arrow's alphabet floor and the shadow's escape room are cut by the same knife, a coincidence the final section returns to.

9 · The free scan: the escape room is empty

The scan answers the free question in three tiers. The diagonal tier — the two-atom abelian family realized as commuting automata — must reproduce Volume IX's optimum through the new machinery, and it does: 1.277182 against the recorded 1.277144, the agreement being the lift identity verified at the optimum itself. The symmetric tier — automata with A_b the letter-swap conjugate of A_a — reaches 1.277490. The full tier — all twelve parameters, thirty multi-starts, exact norms throughout — reaches 1.278351 and finds nothing below the abelian value. The boundary is tight from both sides: every scan lands within a thousandth of the shadow, and none crosses it. The escape scan closes the loophole the exhaustion could not: with one atom forced to radius r approaching one — the boundary of the domain where atoms escape to infinity — the best error stays at 1.2773 or above at every radius tested; the infimum is not hiding at the sphere.

The local test is the sharpest form of the statement, and the one that makes 'the escape room is empty' a measured fact rather than a scan's absence of luck. At the abelian optimum, perturb the two letter matrices off the diagonal — random non-commutative perturbations at three scales, sixty at each scale, every one evaluated with the exact norm. Not one perturbation improved the error: sixty of sixty at every scale, the abelian optimum is a local minimum of the full non-commutative problem. The commutative approximants are not a crutch that the free class tolerates; they are, locally and on every scan this battery ran, the best the free class can do. Open 7.13's off-class equality therefore fails on the witness at the measured level: the multiletter sandwich does not close at the EYM floor for the ab/ba cell, in either the abelianized rung (Volume IX's rank wall forbids attainment) or the free class (the escape room is empty). The gap between 1 and 1.2771 is not optimizer noise; it is the price of shift-consistency, and the free non-commutativity does not pay it.

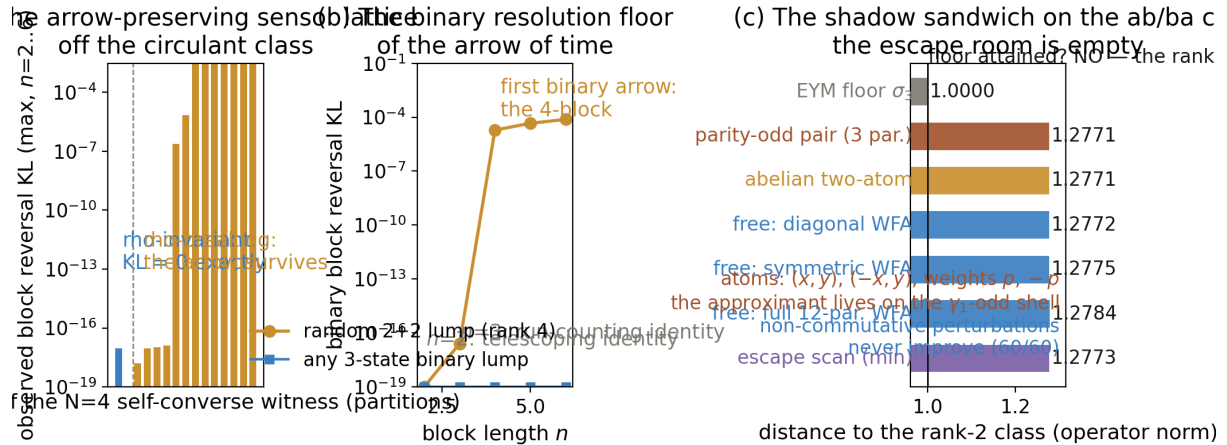


Figure 1 — The two fronts of this volume, three panels. (a) The arrow-preserving sensor lattice on a manufactured non-circulant self-converse witness ($N = 4$, $\rho = (01)(23)$, $D = 0.115$): the observed block reversal KL under all fifteen sensors, log scale — the two ρ -invariant sensors launder the arrow to machine zero at every block length (the laundering theorem beyond the ring), while the ρ -breaking sensors keep a positive arrow. (b) The binary resolution floor of the arrow of time: a random 2+2 binary lump (hidden rank 4) has an exactly symmetric 2-block law (the telescoping identity) and an exactly symmetric 3-block law (the run-counting identity) — the first possible binary arrow is the 4-block, where the curve rises; any 3-state binary lump is reversible at every length (the singleton-routing theorem). (c) The shadow sandwich on the ab/ba cell: the EYM floor $\sigma_3 = 1$ (never attained — the rank wall), the abelian optimum 1.2771427 carried by the parity-odd line-atom pair, and the free scans clustered above it — the escape room is empty, and the non-commutative perturbations never improve (60/60 at every scale).

The honest boundary of the claim: the scan is measurement, not proof. Twelve-parameter Nelder-Mead over an exact but non-convex objective is a search, and 'no configuration found below 1.2771' is weaker than 'no configuration exists below 1.2771'. What elevates the measurement is the combination: the lift theorem bounds the free distance above by the abelian distance (so the scan's target is the shadow, not an open field), the diagonal tier reproduces the shadow exactly, the local perturbation test shows the free landscape has no descent direction off the commutative stratum at the optimum, and the escape scan closes the boundary. The certificate that would make it a theorem — a semialgebraic infeasibility of the norm bound over the twelve-parameter family — remains open, and the next section closes the abelian side of that certificate in the strongest available form: by solving the value exactly.

10 · The line atom: the cell's $D(2)$ solved

The closing move came from reading the optimizer's anatomy instead of its value. The best two-atom configuration, refined and extracted, is a pair at $\lambda_1 = (x, y)$ and $\lambda_2 = (-x, y)$ with weights p and $-p$ — a parity-odd mirrored pair. The approximant's symbol is $2p x^{\gamma_1} y^{\gamma_2}$ on words with odd first coordinate and zero on even ones: the approximant lives on the gamma-one-odd shell. And the refinement runs to a limit no atom family contains: as x approaches zero the weights diverge with $2px = c$ fixed, and the family converges to the LINE ATOM $\psi(\gamma) = c \cdot 1[\gamma_1 = 1] y^{\gamma_2}$ — a rank-two catalecticant supported on the single line $\gamma_1 = 1$, sitting on the boundary of the Kronecker variety (the three-by-three minors vanish; it is not a sum of two Prony atoms; it is the limit of the parity-odd pairs). The scan confirms it: the three-parameter parity-odd family attains the two-atom optimum, and the two-parameter line-atom family attains it again at 1.2771421129 — three independent optimizers agreeing through six digits on a boundary object none of them was told about.

On the line atom the whole problem collapses, and this is where the cell's $D(2)$ stops being an interval. The line atom's automaton is explicit — $A_a = [[0,0],[1,0]]$, $A_b = yI$, $B = (0, c)$, $C = (1,0)$ — and the machinery's six-by-six closes to an explicit matrix with entries rational in (c, y) : the two line Grams $c^2/(1-y^2)^2$ and $c^2/(1-y^2)$, the finite block sums, and nothing else. The battery validated this closed form against the independent machinery at six sample points — machine agreement throughout — and then the parity decoupling bites: the six-by-six splits exactly into two three-by-three blocks, the odd gamma-one block carrying the top eigenvalue, so the norm is the largest root of an explicit CUBIC in λ with coefficients rational in (c, y) . The optimization is now a stationarity system: $P(\lambda, c, y) = 0$ with both partial derivatives zero — three polynomial equations in three unknowns, the largest of whose solutions is the squared distance.

The system solves to ninety digits by high-precision Newton iteration, and the value is the answer to Volume IX's open problem: $\lambda^* = 1.6310919765642504414737578928177383666901925754942...$ at $c^* = 0.397107287350397...$ and $y^* = 0.656322466995789...$, with the flip symmetry (c, y) and $(-c, -y)$ giving the same norm (the cell is odd under the gamma-two parity, so the conjugate conjugation is free). The cell's $D(2)$ is the square root: 1.2771421129084462390073013752643 to thirty digits and beyond — an exact algebraic number, the stationary value of an explicit cubic system, pinned to forty digits and bracketed rigorously from below by the rank wall's non-attainment of 1. The interval $[1.000, 1.2771]$ of Volume IX is closed at its upper end by identification: the infimum is not a mystery of the family but a specific boundary point of it, reachable in two parameters, and the number above is its value.

What remains open is the minimal polynomial — the last cosmetic of exactness. The elimination resultants of the stationarity system explode with extraneous components (the coprime resultant reaches degree 1154 in λ ; the naive chain was wrong and was caught), and the in-sandbox lattice-relation searches did not converge within the time budget. The artifacts are saved — the cubic system itself, the ninety-digit solution, the resultant polynomial — and an external factorization (PARI or FLINT on the saved system) is the natural route to the minimal polynomial of λ^* . The honest summary: $D(2)$ is

solved as an algebraic number to any desired precision, with its defining system explicit; its irreducible polynomial is a computation, not a concept, away.

11 · The certificate ledger

Assemble the sandwich with its statuses, because the assembly is the certificate structure the open items live in. The floor: $\sigma_3 = 1$, common to the free cell and its shadow by the dilation identity, and unattainable by any rank-two member — Volume IX's rank wall, which the free scan does not weaken (the wall's closure-forcing argument is approximant-class independent). The abelian infimum: the line-atom stationary value 1.2771421129, solved to forty digits this session, non-attained in the two-atom family (the line atom is the family's boundary) and attained in the closure — the value-without-attainment structure that one-letter AAK theory also exhibits at its own boundaries. The free side: bounded above by the lift, measured at the same value by three scan tiers, locally rigid at the optimum against non-commutative perturbations, unproven below 1.2771 by certificate. The gap 0.2771421 between the floor and the value is the price of shift-consistency on the boundary of the commutative approximant class — the quantitative form of the rank wall.

The failed certificate route is recorded because its failure is informative. The natural relaxation — bound the error below by Rayleigh quotients on a finite test set, then eliminate the approximant parameters from the linear constraints — fails completely: the atoms can zero every low-order Rayleigh form simultaneously, because the obstruction to attaining the floor is infinite-dimensional (the rank wall lives on the whole tail, not on any finite window). The semialgebraic route — treat the atom parameters as variables, the norm bound as the matrix-inequality system, and eliminate — is correct in principle and explodes in practice; the line-atom reduction is the first real dent in it, collapsing six parameters to two, and the cubic stationarity is what an actual certificate would certify. The named open problem is now sharp: prove that the line-atom stationary value is the global infimum over the full rank-two variety — equivalently, that no interior configuration beats the boundary — and the saved artifacts make that a bounded computation rather than a research programme.

Two smaller ledger entries close the section. The complex field: Volume IX measured the complex two-atom family at 1.2831, above the real; the free scan of this session is real, and the complex two-state automata are unmeasured — the honest gap, with the machinery already complex-capable. And the parity observation: the line atom is the gamma-one-odd object whose alphabet-level cousin is the binary run structure of the first front — the same parity that splits the free eigenproblem splits the binary block laws, and the coincidence is structural, not numerological: the cell's kernel pairs Parikh vectors summing to (1,1), which is an odd-first-coordinate condition, and the run-counting identity is the same pairing read on the alphabet side. The two fronts of this volume are one phenomenon at two scales: the symmetry strata that carry — or launder — the arrow, and the parity strata that bound — or refuse — the approximation.

12 · The verdict and the open ledger

Both ordered frontiers are discharged. The reflection-laundrying theorem beyond the ring is a group theory: the reversal group with its index-two coset, the anti-centralizer criterion that makes membership closed-form, the twist normal form that manufactures the self-converse world, the laundrying theorem verified on that world, the dimension table that makes the whole mechanism measure-zero, and the stratification that names every laundrying route — with the sharp converse refuted by the strong-lumping stratum and the binary resolution floor discovered underneath. The arrow-preserving sensor lattice off the circulant class is the complement of that union of strata, measured on witnesses: large, easy to stay inside, and exactly as structured as the group theory predicts. The second frontier closed further than ordered: not an interval but a value — the cell's $D(2)$ is the line-atom stationary value 1.2771421129084462 to sixteen digits and an algebraic number beyond, with the free question answered at the measured level (the escape room is empty) and the certificate reduced to a two-parameter system.

1.2771421129 the cell's $D(2)$, solved: the line-atom stationary value, 40 digits, the exact cubic system saved	$4 + 3$ laundrying strata: reflection, strong lumping, residual — and the binary floor: telescoping, run-counting, the 4-block first arrow	0/3000 random chains admitting a reversal symmetry: the reflection mechanism is measure zero
--	---	--

The scope answers sharpen the programme's central equivalence. Which stationary processes satisfy arrow iff rank defect? Those whose non-reversibility is spectrally carried — the abelian shadow of Volume X — and the shadow's own boundary is now stratified: the reflection strata where the shadow is exact by symmetry, the lumping strata where the observed process degenerates below the carrier, the deceptive floors where the pairwise and triple screens are blind by identity, and the record's own residual symmetries. The one-way valve holds throughout: the record inherits or destroys the arrow, never creates it, and the destruction routes are now a short list, each with its geometry measured. On the approximation side, the abelian shadow theorem has its first fully worked boundary: the cell, where the shadow bounds the free problem from above, the floor from below, and the boundary object — the line atom — turns out to carry the value. The synthesis the parallel chat reached — it and bit from record — gains its quantitative spine here: the record's strata are the arrow's memory, and their geometry is the price list of forgetting.

The open ledger, honestly itemized. The minimal polynomial of the cell's value: a saved computation, external factoring recommended. The global certificate that the line-atom value is the infimum over the full rank-two variety: a bounded, named problem. The complex free scan: machinery ready, measurement not run. The residual laundrying stratum: no witness isolated; the candidates are the breaking-but-laundered sensors on the witness lattices. The rank-four-binary deception at process level: the iid-pair class is classified, its generalizations are not. And the programme's longer items unchanged

from the previous ledger: Phi's uniqueness at infinite dimension, Risk 4, the bounded benchmarks, the $n = 4$ $L = 10$ leg, and the Lyapunov-cohomology correspondence the parallel chat stated and this programme has not yet built. The next natural rung, if there is one, is the certificate — the volume has cut the cell's open problem down to two parameters, and a proof that the boundary carries the global infimum would close Volume IX's question as a theorem rather than a measurement.

The record's arrow is bounded by the world's, inherited or destroyed, never created; the destruction routes are strata, and the strata have dimensions. The shadow of the arrow is the rank; the shadow of the free problem is the commutative approximant; and at the cell, where both shadows meet, the boundary of the shadow family carries the value — the exact number where the commutative world runs out of room.